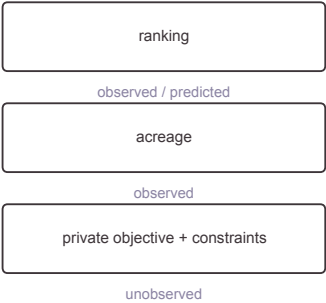


Ordinal rankings do not identify cardinal allocations

a Which objects are needed for each allocation claim?

|                    |              |                 |            |              |              |                |
|--------------------|--------------|-----------------|------------|--------------|--------------|----------------|
| Rank disagreement  | •            | —               | —          | —            | —            | —              |
| Possible reversal  | •            | •               | •          | •            | •            | —              |
| Universal reversal | •            | •               | •          | •            | •            | —              |
| Selected reversal  | •            | •               | •          | •            | •            | •              |
|                    | Ordinal rank | Cardinal payoff | Joint risk | Feasible set | Optimal face | Selection rule |

b Observability is not identification



c The full optimal face, not one solver output, determines set-valued reversal

